

SUMMARY

Data Engineer with 5 Years of experience specializing in Python, SQL, cloud platforms, and scalable data pipeline architectures. Proficient in ETL processes, real-time data streaming with Apache Kafka and Spark, and leveraging AWS services to optimize data workflows and enable data-driven decision-making.

SKILLS

- Programming Languages:** Python, C, SQL, R
- ETL Tools:** Azure Data Factory, DBT, AWS Glue, Apache Airflow, Databricks
- Data Pipelines:** PySpark, Spark SQL, ETL/ELT, Spark Streaming, RDD Transformations
- Data Warehousing:** Snowflake, Redshift, Azure SQL Data Warehouse, Google BigQuery
- Big Data Ecosystem:** Cloudera Hadoop, HDFS, Hive, Parquet, YARN, Aapache Kafka, Apache Spark
- Azure:** Azure Synapse Analytics, Azure Data Lake, Azure Blob Storage, Azure HDInsight, Azure Cosmos DB, Azure SQL Database, AKS
- AWS:** S3, EC2, RDS, Athena, EMR, Lambda
- Google Cloud Platform:** BigQuery, Cloud Storage, Dataflow, Pub/sub
- Databases:** MongoDB, PostgreSQL, MySQL, NoSQL, Oracle
- DevOps Tools:** Jenkins, Terraform, Docker, Kubernetes
- Analytics & Visualization:** Power BI, Tableau, Looker, SQL Server Reporting Services (SSRS), Google Data Studio, Matplotlib, Seaborn
- Machine Learning:** Scikit-learn, TensorFlow, PyTorch, Keras
- Version Control & Collaboration:** Git, GitHub, Bitbucket, JIRA

EXPERIENCE

Data Engineer

May 2024 – Present

Metlife

USA

- Deployed 20+ Azure Functions using Python in Azure HDInsight, enabling real-time data processing and lessen runtime by 30%.
- Designed a Security Framework for fine-grained access control in Azure Blob Storage by integrating Azure Functions, Cosmos DB, and Terraform, upgrading data security by 25%.
- Constructed scalable ETL pipelines in Azure Data Factory, migrating 50+ terabytes of data to Azure Synapse Analytics, and minimizing data loading time by 40%.
- Implemented CI/CD pipelines with Docker, Kubernetes, and Azure DevOps, leveraging Terraform to provision infrastructure, achieving 99.9% deployment reliability.
- Established streaming data pipelines making use of Azure Databricks, Event Hubs, and Hive, decreasing latency by 20% and providing seamless integration with machine learning models.
- Streamlined ETL workflows harnessing Azure Data Factory and Terraform, minimizing manual interventions by 60% and processing millions of records across multiple Azure environments.
- Provisioned Azure infrastructure exploiting Terraform modules, including virtual networks, storage accounts, and AKS clusters, scale back provisioning time by 50%.
- Launched ML models with Azure ML Services, rationalising predictive analytics workflows and boosting operational efficiency by 25% through automated retraining pipelines.
- Enhanced data transformation workflows on Azure HDInsight with Hive and Spark, cutting data errors by 30% and improving real-time analytics performance.
- Managed Terraform state files in Azure Blob Storage, ensuring efficient collaboration and drift detection across teams in multi-region deployments.

Data Warehouse Analyst

July 2021 – July 2023

Cognizant Technology Solutions

India

- Integrated healthcare databases across AWS (RDS, Redshift) and GCP BigQuery, eliminating data inconsistencies by 20% while ensuring HIPAA compliance for all environments.
- Assembled scalable ETL pipelines adopting AWS Glue, processing millions of records from EHR/EMR systems and medical devices into AWS S3 with 99.9% data accuracy.
- Engineered Apache Spark scripts for high-performance aggregation and transformation of large healthcare datasets, augmenting data processing efficiency.
- Automated data workflows with Apache Airflow and Python, refining manual interventions by 80% and optimizing processing times for clinical trial data by 35%.
- Created real-time data ingestion pipelines with AWS Lambda and GCP Cloud Functions, streamlining IoT healthcare data integration into Snowflake, reducing costs by 25%.
- Directed migration of large Hadoop datasets (patient records, genomic data) to AWS S3 and GCP Storage, transforming storage performance by 50% and reducing costs by 20%.
- Applied Protegrity tokenization to secure PII/PHI data, ensuring compliant migration of sensitive healthcare data to cloud environments.
- Deployed containerized microservices on AWS ECS and GCP GKE, enhancing scalability and diminishing deployment time by 40%.
- Utilized AWS Glue and GCP Dataflow for data cleansing and transformation, advancing dataset quality by 15% for predictive healthcare analytics.
- Developed Tableau dashboards and performed EDA to visualize hospital readmissions, patient satisfaction, and provider performance metrics.

Data Analyst

July 2020 – June 2021

Experion Technologies

India

- Innovated ETL pipelines employing Apache Spark and Hive, consolidate data processing times for automotive sensor data by 40%, enabling predictive maintenance insights.
- Formulated a real-time data platform on Azure Synapse Analytics, lowering unplanned maintenance by 30% across 100,000+ vehicles.
- Led Hadoop and Elasticsearch integration, allowing real-time search on 2PB of structured and unstructured automotive data, fine-tuning response times by 50%.
- Simplified data ingestion and transformation with Informatica and Talend, elevating data delivery timeliness by 20% and ensuring accuracy in critical automotive datasets.
- Adopted Docker-based containerized environments and orchestrated deployments with Kubernetes, enhancing scalability and streamlining production releases.
- Created custom Power BI dashboards to monitor KPIs, empowering automotive teams to make data-driven decisions and boost operational performance.
- Collaborated with IT and engineering teams to optimize big data workflows, trimming project turnaround times by 15% and improving cross-functional efficiency.

EDUCATION

Master of Science in Information Technology

August 2023 – December 2024

University of Cincinnati, Cincinnati, USA

Bachelor of Technology in Electrical and Electronics Engineering

April 2017 – May 2021

V. R. Siddhartha Engineering College, Vijayawada, India

PROJECTS

Real-Time Clickstream Data Processing for Web Analytics:[[Github Link](#)]

Instituted a real-time data pipeline to process web clickstream data to track user behavior and improve website personalization. Data is ingested via Apache Kafka, processed in real-time incorporating Apache Flink, and stored in Google BigQuery. Provided instant insights into user activity, allowing the marketing team to adjust content in real time, leading to a 20% increase in user engagement.

Peer-to-peer Loan Investment Risk Modelling:[[Github Link](#)]

Engineered an ETL pipeline for peer-to-peer loan investment risk modeling, utilizing Apache Kafka, Apache Spark, and AWS services to process and store large datasets. Built machine learning models to predict loan default risks, strengthening investment decision-making. Enforced real-time data analytics and interactive dashboards to optimize loan portfolio performance and reduce defaults by 15%.